

Cressie-Read Power Divergence for Moment-Based Estimation in Machine Learning

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UIUC Econometrics Brownbag Seminar

1. Introduction

Topic

- Finite-sample considerations are increasingly central to modern approaches to estimation and inference.
 - ▶ In some empirical settings, data collection is costly or constrained, leading to limited sample sizes.
 - ▶ In high-dimensional settings ($n \ll p$), the effective information per parameter is low, so the problem behaves like a small-sample environment (see, e.g., Huber, 1973).

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- In finite samples, GMM estimators may perform poorly (see, e.g., Newey and Smith, 2004).
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- In finite samples, GMM estimators may perform poorly (see, e.g., Newey and Smith, 2004).
 - ▶ Estimation of the optimal weighting matrix can be noisy and unstable, especially when a small number of observations produce large moment residuals.
- This perspective highlights a key advantage of Cressie–Read power divergence (CRPD) estimators (Cressie and Read, 1984) in moment-based estimation.
 - ▶ Dual to generalized empirical likelihood (GEL) estimator, which is known to perform better than GMM in finite samples.
e.g., see Owen, 1988; Qin and Lawless, 1994; Imbens, 1997; Hansen, Heaton, and Yaron, 1996; Kitamura and Stutzer, 1997; Imbens, Spady, and Johnson, 1998; Newey and Smith 2004; Schennach, 2007.

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 - ▶ CRPD compares the distance between two probability distributions.
 - ▶ γ governs the curvature of the CRPD divergence.
 - ★ $\gamma > 0$: amplifies large probability discrepancies.
 - ★ $\gamma = 0$: yields logarithmic sensitivity, providing a smooth and stable benchmark.
 - ★ $\gamma < 0$: compresses extreme probability discrepancies.



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 - ▶ Popular choices
 - ★ $\gamma = -1$: empirical likelihood (EL)
 - ★ $\gamma = 0$: exponential tilting (ET)
- γ is typically chosen *a priori* by the researcher. Why?
 - ▶ Researchers often select familiar benchmark estimators (e.g., EL, ET, continuous updating), corresponding to specific values of γ .
 - ▶ The resulting estimators are first-order asymptotically equivalent under correct specification (see, e.g., Newey and Smith, 2004).

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Motivation: different penalty on outliers

- Let $m = 15$ be the number of outliers.

n	Variable	mean	sd	Q1	Q2	Q3	max
100	x	0.097	3.926	-0.543	-0.059	0.589	10.000
100	y	2.185	39.180	-0.127	1.048	2.244	121.000

- ▶ Data with outliers.
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- Let the objective loss function be

$$Q_n(\beta; a) = \frac{1}{n} \sum_{i=1}^n |y_i - \theta_0 - \theta_1 x_i|^a, \quad a > 0.$$

- ▶ $a = 1$: LAD (Least Absolute Deviations):
 - ★ Minimizes absolute residuals.
 - ★ Estimates the conditional median.
 - ★ More robust to outliers than OLS (linear penalty).
- ▶ $a = 2$: OLS

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Motivation: different penalty on outliers

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 - ▶ In practice, we determine a via cross-validation.
- The cross-validation results are summarized as follows.

a	fold	mse	average mse
1	1	1,608.209	1,565.830
	2	1,535.575	
	3	500.282	
	4	2,000.896	
	5	2,184.190	
2	1	2,038.409	1,800.426
	2	1,728.935	
	3	530.515	
	4	2,031.743	
	5	2,672.531	

- ▶ i.e., $a = 1$ gives the lowest MSE among $a \in \{1, 2\}$.

1. Introduction

Motivation: different penalty on outliers

- Estimation results

Parameter	LAD ($a = 1$)	OLS ($a = 2$)
$\theta_0^0 = 1$	0.9519 (0.0494) [0.0481]	2.0553 (3.9036) [1.0553]
$\theta_1^0 = 2$	1.8976 (0.0126) [0.1024]	1.3382 (0.9989) [0.6618]

- ▶ With $m/n = 0.15$, OLS is more biased and larger variance than LAD.
- ▶ This exercise implies that the optimal loss function, governed by the hyperparameter a , depends on the features of the data.

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Research question

- **Observation**

- ▶ In the presence of irregular data (e.g., outliers), applying different penalties can improve robustness.
- ▶ This is particularly important in finite samples, where a few influential observations can distort moment conditions and estimation accuracy.

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Research question

● Observation

- ▶ In the presence of irregular data (e.g., outliers), applying different penalties can improve robustness.
- ▶ This is particularly important in finite samples, where a few influential observations can distort moment conditions and estimation accuracy.

● Questions

- ▶ If γ shapes the curvature of the penalty, its role becomes particularly important in small samples.
- ▶ The generalized empirical likelihood (GEL) estimator—dual to CRPD—was motivated by finite-sample improvements over GMM, suggesting that first-order equivalence alone may be insufficient.
- ▶ Is γ merely an indexing parameter, or does it influence the curvature of the objective and the weighting mechanism in CRPD beyond serving as a conventional index parameter?

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Contribution

We show that the power parameter γ should be viewed as a *learning hyperparameter* in the sense of modern machine learning, rather than selected arbitrarily.

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 - ▶ γ controls the curvature of the penalty used to enforce moment conditions through empirical reweighting.

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Contribution

We show that the power parameter γ should be viewed as a *learning hyperparameter* in the sense of modern machine learning, rather than selected arbitrarily.

1. γ determines the loss function and thus governs the geometry of the learning objective.
 - ▶ γ controls the curvature of the penalty used to enforce moment conditions through empirical reweighting.
2. γ plays a substantive role in the *second*-order behavior of both the structural estimator and the associated Lagrange multipliers.
 - ▶ Second-order terms determine learning performance (see, e.g., Rothenberg (1984); Newey and Smith (2004); Huber and Ronchetti (2009)).

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Related literature

1. Robust M-estimation

- **Huber (1964, AoS); Huber and Ronchetti (2009)**

- ▶ Robustness achieved through loss design.
- ▶ Curvature of the loss controls the influence of extreme observations.
- ▶ Tuning constants govern the finite-sample robustness–efficiency trade-off.
- ▶ These constants are not structural parameters of the DGP.

- **Our framework**

- ▶ Extends the loss-design perspective to moment-based estimation.
- ▶ The CRPD power parameter controls the curvature of a divergence-based learning objective.
- ▶ Curvature governs empirical reweighting and constraint enforcement.
- ▶ Like robust M-estimation, γ is not identified by population moments, but shapes finite-sample behavior.

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Related literature

2. Higher-order asymptotic theory

- **Newey and Smith (2004, *Econometrica*)**

- ▶ Develop higher-order asymptotics to compare the finite-sample performance of GMM and generalized empirical likelihood (GEL).
- ▶ The power parameter primarily serves to index different members of the GEL family.
- ▶ First-order behavior of the Lagrange multipliers is characterized; at this order, all CRPD members are asymptotically equivalent.

- **Our framework**

- ▶ Build on higher-order expansion techniques but focus on a different object.
- ▶ Treat γ as a parameter shaping the geometry of the CRPD objective rather than merely indexing estimators.
- ▶ Derive second-order behavior of the Lagrange multipliers, where curvature effects emerge and γ enters nontrivially.
- ▶ Show how γ influences finite-sample constraint enforcement and robustness—channels that are not visible in first-order theory.

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Related literature

3. Mixture of CRPD estimators

- **Schennach (2007, AoS)**

- ▶ Develops exponentially tilted empirical likelihood (ETEL).
- ▶ Bridges EL and ET via alternative power parameter choices.
- ▶ Focuses on robustness under moment misspecification.
- ▶ Shows that EL and ET, though first-order equivalent under correct specification, can differ under misspecification.

- **Our framework**

- ▶ Maintain correct moment specification.
- ▶ Treat the power parameter as a continuum-valued learning hyperparameter.
- ▶ Analyze how the power parameter works as a divergence curvature that affects second-order behavior.
- ▶ Show that EL and ET can differ in finite samples even under correct specification.
- ▶ Differences arise through second-order effects driven by the power parameter.

1. Introduction

Recap: hyperparameter in machine learning

- **Hyperparameter**

- ▶ A tuning parameter set outside the main estimation problem.
- ▶ Governs the learning environment.

- **Role in estimation**

- ▶ Enters the objective externally (e.g., penalty weight, learning rate, tree depth).
- ▶ Typically selected via cross-validation or other data-driven criteria.

- **Connection to finite-sample behavior**

- ▶ Does not affect identification or first-order consistency.
- ▶ Shapes the curvature of the objective function.
- ▶ Therefore influences finite-sample bias, variance, and robustness.
- ▶ Asymptotically negligible, but economically meaningful in small samples.
 - ★ Note. Small sample includes high-dimensional settings, where the effective sample size per parameter is limited despite a large nominal dataset. (e.g., see Huber, 1973).

1. Introduction

Outline

1. The role of γ in the Cressie-Read Power Divergence (CRPD)
 - ▶ γ and the curvature of loss functions
 - ▶ Moment-based CRPD estimation
2. Monte Carlo simulations
3. Empirical application
 - ▶ Data from Owen (1988) ($n = 22$)
4. Conclusion

The Role of γ in Loss Curvature
in the Cressie–Read Power Divergence (CRPD)

2. Cressie-Read Power Divergence (CRPD)

Definition 1 (Cressie-Read Power Divergence)

Consider two discrete distributions with common support $\mathbf{p} = (p_1, \dots, p_n)$ and $\mathbf{q} = (q_1, \dots, q_n)$, where n is the number of observations. Rewrite the CRPD criterion as

$$I_\gamma(\mathbf{p}, \mathbf{q}) = \frac{1}{\gamma(\gamma + 1)} \sum_{i=1}^n p_i \left[\left(\frac{p_i}{q_i} \right)^\gamma - 1 \right], \quad (1)$$

where $\gamma (\neq 0, -1)$ is a parameter which indexes a one-parameter family of divergence measures.

- We consider the *uniform* distribution over the data points as the reference measure, $\mathbf{q} := \mathbf{i}/n$.
 - ▶ This choice provides a consistent empirical approximation to the unknown true distribution in the absence of prior information.
- The CRPD family forms a continuum indexed by the power parameter γ .

2. Cressie-Read Power Divergence (CRPD)

- Observe that γ determines a *loss* function. For example,

$$l_{\gamma}(\boldsymbol{\pi}, \mathbf{i}/n)|_{\gamma=1} = \frac{1}{2} \sum_{i=1}^n \pi_i [n\pi_i - 1] = \frac{1}{2} \left[n \sum_{i=1}^n \pi_i^2 - \underbrace{\sum_{i=1}^n \pi_i}_{=1} \right] = \frac{1}{2} \left[n \sum_{i=1}^n \left(\pi_i - \frac{1}{n} \right)^2 \right].$$

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- Different values of γ correspond to different divergences and therefore different loss curvature on the probability simplex.
 - ▶ $\gamma = -1$ (**EL**):
 - ★ Log-barrier geometry prevents weights from approaching zero.
 - ★ Produces more dispersed weights to satisfy the moment conditions.
 - ▶ $\gamma = 0$ (**ET**):
 - ★ Corresponds to the Kullback–Leibler divergence.
 - ★ Penalizes deviations from the uniform distribution smoothly.
 - ★ Weights tend to remain close to $1/n$ when the moment restrictions are approximately satisfied.
 - ▶ $\gamma = 1$ (**Pearson χ^2**):
 - ★ Quadratic geometry.
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 - ▶ $\gamma = 1$ (**Pearson χ^2**):
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 - ★ Penalizes deviations symmetrically around the uniform benchmark.
 - ▶ Thus γ controls the **curvature of the loss**, which determines how aggressively the estimator reweights the data.

γ value	Core loss term	Geometry	Interpretation
$\gamma = -1$ (EL)	$-\sum_i \log(n\pi_i)$	Log barrier	Empirical likelihood. The objective diverges as $\pi_i \downarrow 0$, strongly discouraging vanishing weights and enforcing interior solutions.
$\gamma = 0$ (ET)	$\sum_i \pi_i \log(n\pi_i)$	Logarithmic	Exponential tilting. Corresponds to the Kullback–Leibler divergence from the uniform distribution; weights are smoothly reweighted and remain close to $1/n$ when the moment conditions are nearly satisfied.
$\gamma = 1$ (Pearson χ^2)	$\sum_i \pi_i^2$	Quadratic	Pearson χ^2 divergence. Equivalent to the squared Euclidean distance from the uniform distribution on the probability simplex; in moment space it corresponds to a quadratic (Hotelling-type) distance.

Moment-based CRPD estimation

3. Moment-based CRPD estimation

- We consider CRPD estimation with moment conditions.
- Key takeaway:
 - ▶ the power parameter γ does *not* affect first-order asymptotics,
 - ▶ but enters the *second-order* behavior of the structural parameter and Lagrange multipliers,
 - ▶ thus affecting finite-sample performance.

Outline of the proof

0. Assumptions

- ▶ regularity conditions for identification, smoothness, compactness, and local feasibility
- ▶ ensure that the profiled CRPD problem is well behaved

1. Existence and uniqueness of multipliers

- ▶ for each θ near θ_0 , show that $(\hat{\lambda}(\theta), \hat{\delta}(\theta))$ exists and is unique
- ▶ this guarantees that the profiled criterion $L_n(\theta)$ is well defined

2. Consistency of the estimator and multipliers

- ▶ uniform convergence of the profiled objective implies $\hat{\theta} \xrightarrow{P} \theta_0$
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3. First-order asymptotics of $\hat{\lambda}$

- ▶ write a mean-value expansion of the weights.
- ▶ plug the mean-value expansion of the weights to the moment constraint and consistency of $\hat{\delta}$ yields

$$\sqrt{n}\hat{\lambda} \xrightarrow{d} N(0, \Omega_0^{-1}),$$

where

$$\Omega_0 := E[g(Z, \theta_0)g(Z, \theta_0)']$$

is the population covariance matrix of the moments.

- ▶ plugging the mean-value expansion of the weights to the probability constraint and $\hat{\lambda} = O_p(n^{-1/2})$ yield $\hat{\delta} = O_p(n^{-1})$.

Outline of the proof (continued)

4. Second-order expansion of the weights

- ▶ define the local term

$$t_i := (\hat{\delta} - \delta_0) + \hat{\lambda}' g_i(\theta) = O_p(n^{-1/2})$$

using $\hat{\lambda}$ and $\hat{\delta}$.

- ▶ since $t_i = o_p(1)$, expand around $t = 0$:

$$(1 - \gamma t_i)^{1/\gamma} = 1 - t_i + \frac{1 - \gamma}{2} t_i^2 + \dots$$

- ▶ γ enters through this curvature expansion

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5. Limiting distribution of $\hat{\delta}$

- ▶ substitute the second-order weight expansion into the probability constraint.
- ▶ subsequently substituting the leading term of $\hat{\lambda}$ gives

$$n(\hat{\delta} - \delta_0) \xrightarrow{d} -\frac{\gamma + 1}{2} \chi_q^2.$$

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6. Second-order expansion of $\hat{\lambda}$

- ▶ substitute the second-order weight expansion into the moment constraint
- ▶ obtain

$$\hat{\lambda} = \hat{\Omega}(\theta_0)^{-1} \bar{g}(\theta_0) + \frac{1}{n} B_{\lambda,n}(\gamma) + o_p(n^{-1}),$$

where $B_{\lambda,n}(\gamma) = O_p(1)$

Outline of the proof (continued)

7. Second-order expansion of $\hat{\theta}$

- ▶ use the (envelope) FOC for θ :

$$\left. \frac{\partial \mathcal{L}}{\partial \theta} \right|_{\hat{\pi}(\theta), \hat{\lambda}(\theta), \hat{\delta}(\theta)} \stackrel{\text{FOC}}{=} 0$$

- ▶ plug the Taylor expansion around θ_0 of

$$\bar{G}(\hat{\theta}) = \frac{1}{n} \sum_{i=1}^n G_i(\theta), \quad G_i(\theta) = \frac{\partial g_i(\theta)}{\partial \theta},$$

together with the second-order expansion of $\hat{\lambda}$ into the envelope FOC.

- ▶ solve for

$$\hat{\theta} - \theta_0 = (G_0' \Omega_0^{-1} G_0)^{-1} G_0' \Omega_0^{-1} \bar{g}(\theta_0) + \frac{1}{n} B_{\theta,n}(\gamma) + o_p(n^{-1})$$

where

$$G_0 := E \left[\frac{\partial g(Z, \theta_0)}{\partial \theta} \right]$$

is the population Jacobian of the moments, and $B_{\theta,n}(\gamma) = O_p(1)$.

Monte Carlo Simulations

4. Simulation

Setup

- **Design**

- ▶ 1,000 iterations.
- ▶ Sample sizes: $n \in \{25, 50\}$.

- **Data generating process**

- ▶ $Z_i \sim t_\nu$, $\nu \in \{5, 15\}$ and normal.
 - ★ Smaller $\nu \Rightarrow$ heavier tails.
 - ★ Larger $\nu \Rightarrow$ approximately normal.
- ▶ True parameter: $\theta_0 = (\mu_0, \sigma^2) = (0, 1)$.

4. Simulation

Setup

- **Moment structure**

- ▶ Baseline moments

$$E[X - \mu] = 0, \quad E[(X - \mu)^2 - \sigma^2] = 0.$$

- ▶ Issue: with exact moments, CRPD collapses to the sample mean ($\hat{\theta} = \bar{X}$, uniform weights, $\lambda = 0$).
- ▶ To avoid this trivial solution, we impose a symmetry moment

$$E[(X - \mu)^3] = 0.$$

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- ★ This induces non-uniform weighting across observations.



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- **Search procedure**

- ▶ $\gamma \in [-1, 1]$ with step 0.25 (for illustration).
- ▶ θ searched over a neighborhood containing the true value.
- ▶ Adaptive mesh

$$\Delta_{\theta} = \frac{\hat{\sigma}_X}{c_{\theta} \sqrt{n}}.$$

- ★ Shrinks at the $\sqrt{n}$ rate: Matches the local scale $\hat{\theta} - \theta_0 = O_p(n^{-1/2})$.
- ▶ For each CRPD estimation, γ is fixed and profile w.r.t. θ . For each θ , we solve multipliers (λ, δ) and weights π .

4. Simulation

Finite sample metrics ($n = 25$)

DGP	γ	Second-order measure			First-order measure		
		Bias	MSE	Coverage distortion	emp_sd	mean_se	se_ratio
t_5	-1.00	-0.0174	0.0923	-0.1090	0.3034	0.2192	1.3841
	-0.75	-0.0354	0.1089	-0.1374	0.3283	0.2182	1.5044
	-0.50	-0.2125	0.2776	-0.2806	0.4823	0.2309	2.0893
	-0.25	-0.4166	0.4440	-0.4075	0.5203	0.2473	2.1039
	0.00	-0.1347	0.2138	-0.1860	0.4425	0.2317	1.9100
	0.25	-0.0007	0.1109	-0.0821	0.3331	0.2221	1.4999
	0.50	-0.0205	0.1412	-0.1155	0.3754	0.2215	1.6947
	0.75	-0.0436	0.1561	-0.1388	0.3929	0.2241	1.7533
	1.00	-0.0685	0.1822	-0.1764	0.4215	0.2235	1.8859
Normal	-1.00	-0.0180	0.0540	-0.1210	0.2317	0.1703	1.3608
	-0.75	-0.0416	0.0693	-0.1350	0.2602	0.1715	1.5172
	-0.50	-0.2743	0.2592	-0.4060	0.4291	0.1884	2.2778
	-0.25	-0.4780	0.4040	-0.5550	0.4192	0.2079	2.0162
	0.00	-0.1594	0.1771	-0.2230	0.3896	0.1864	2.0904
	0.25	-0.0003	0.0506	-0.1040	0.2250	0.1732	1.2992
	0.50	0.0032	0.0536	-0.1110	0.2317	0.1720	1.3465
	0.75	0.0036	0.0529	-0.1130	0.2300	0.1711	1.3439
	1.00	0.0018	0.0587	-0.1140	0.2423	0.1714	1.4141

4. Simulation

Finite sample metrics ($n = 50$)

DGP	γ	Second-order measure			First-order measure		
		Bias	MSE	Coverage distortion	emp_sd	mean_se	se_ratio
t_5	-1.00	-0.0020	0.0363	-0.0465	0.1906	0.1638	1.1641
	-0.75	-0.0034	0.0360	-0.0373	0.1898	0.1638	1.1584
	-0.50	-0.0730	0.1130	-0.1100	0.3283	0.1677	1.9574
	-0.25	-0.3466	0.3863	-0.3460	0.5162	0.1826	2.8265
	0.00	-0.0412	0.0712	-0.0610	0.2638	0.1664	1.5855
	0.25	-0.0267	0.0853	-0.0870	0.2910	0.1669	1.7434
	0.50	-0.0380	0.1078	-0.0960	0.3263	0.1680	1.9421
	0.75	-0.0427	0.1155	-0.1232	0.3374	0.1680	2.0085
	1.00	-0.0565	0.1334	-0.1680	0.3611	0.1697	2.1279
Normal	-1.00	-0.0060	0.0219	-0.0390	0.1478	0.1311	1.1275
	-0.75	-0.0071	0.0257	-0.0410	0.1602	0.1314	1.2189
	-0.50	-0.1240	0.1345	-0.1870	0.3453	0.1369	2.5218
	-0.25	-0.4980	0.4608	-0.5750	0.4615	0.1575	2.9302
	0.00	-0.0493	0.0653	-0.0940	0.2509	0.1345	1.8646
	0.25	-0.0017	0.0222	-0.0470	0.1491	0.1313	1.1355
	0.50	-0.0021	0.0219	-0.0350	0.1480	0.1321	1.1203
	0.75	0.0013	0.0244	-0.0440	0.1563	0.1322	1.1820
	1.00	-0.0089	0.0277	-0.0530	0.1661	0.1328	1.2514

4. Simulation

Results ($n = 25$)

DGP	γ	$\hat{\theta}$	Lagrange multipliers					Minimum	Q1	Median	π		
			λ_1	λ_2	λ_3	δ	Mean				Q3	Maximum	
t_5	-1.00	-0.0174 (0.3034)	-0.0648 (1.2644)	-0.0797 (1.3803)	0.1092 (2.0990)	-1.00E+04 (1.42E-05)	0.0208 (0.0122)	0.0337 (0.0051)	0.0374 (0.0043)	0.0400 (3.88E-09)	0.0449 (0.0042)	0.0642 (0.0336)	
	-0.75	-0.0354 (0.3283)	-2.92E+10 (5.15E+11)	-2.01E+10 (3.44E+11)	1.09E+10 (1.85E+11)	-1.70E+09 (3.46E+10)	0.0192 (0.0128)	0.0331 (0.0075)	0.0367 (0.0076)	0.0388 (0.0067)	0.0437 (0.0087)	0.0600 (0.0344)	
	-0.50	-0.2125 (0.4823)	-3.38E+12 (9.04E+13)	-1.91E+12 (4.52E+13)	8.70E+11 (1.78E+13)	-2.47E+11 (6.65E+12)	0.0124 (0.0129)	0.0258 (0.0152)	0.0288 (0.0167)	0.0304 (0.0171)	0.0345 (0.0198)	0.0489 (0.0459)	
	-0.25	-0.4166 (0.5203)	-1.66E+12 (1.53E+13)	-1.87E+12 (1.82E+13)	1.14E+12 (1.09E+13)	-2.04E+11 (2.01E+12)	0.0088 (0.0126)	0.0178 (0.0177)	0.0197 (0.0194)	0.0211 (0.0200)	0.0231 (0.0227)	0.0398 (0.0723)	
	0.00	-0.1347 (0.4425)	4.51E+01 (3.40E+02)	-8.70E+01 (5.26E+02)	2.70E+01 (1.51E+02)	-8.82E+01 (-6.68E+02)	0.0155 (0.0138)	0.0296 (0.0140)	0.0326 (0.0151)	3.30E+74 (-1.04E+76)	0.0370 (0.0172)	8.24E+75 (2.61E+77)	
	0.25	-0.0007 (0.3331)	0.0070 (0.4426)	-0.0190 (0.0952)	-0.0024 (0.1307)	-0.8618 (0.1263)	0.0164 (0.0137)	0.0352 (0.0053)	0.0388 (0.0041)	0.0400 (5.43E-10)	0.0446 (0.0038)	0.0598 (0.0354)	
	0.50	-0.0205 (0.3754)	0.0271 (0.4858)	-0.0188 (0.0874)	-0.0036 (0.1254)	-0.7522 (0.1810)	0.0160 (0.0139)	0.0347 (0.0064)	0.0385 (0.0045)	0.0400 (2.24E-10)	0.0447 (0.0038)	0.0612 (0.0367)	
	0.75	-0.0436 (0.3929)	0.0498 (0.5223)	-0.0173 (0.0872)	-0.0040 (0.1217)	-0.6803 (0.2200)	0.0157 (0.0141)	0.0341 (0.0075)	0.0385 (0.0044)	0.0400 (1.26E-10)	0.0452 (0.0042)	0.0610 (0.0326)	
	1.00	-0.0685 (0.4215)	0.0919 (0.6059)	-0.0200 (0.0800)	-0.0051 (0.1294)	-0.6494 (0.2983)	0.0157 (0.0142)	0.0331 (0.0084)	0.0383 (0.0045)	0.0400 (1.44E-10)	0.0460 (0.0052)	0.0627 (0.0327)	
Normal	-1.00	-0.0180 (0.2317)	-0.0522 (0.6503)	-0.0309 (0.3998)	0.0812 (0.7750)	-1.00E+04 (1.16E-05)	0.0270 (0.0096)	0.0339 (0.0049)	0.0376 (0.0038)	0.0400 (3.79E-09)	0.0447 (0.0035)	0.0588 (0.0295)	
	-0.75	-0.0416 (0.2602)	-2.82E+09 (6.35E+10)	-2.47E+09 (5.11E+10)	1.64E+09 (3.00E+10)	-1.23E+08 (2.83E+09)	0.0252 (0.0111)	0.0327 (0.0082)	0.0363 (0.0086)	0.0382 (0.0083)	0.0429 (0.0100)	0.0540 (0.0314)	
	-0.50	-0.2743 (0.4291)	-1.99E+12 (3.77E+13)	-2.46E+12 (4.33E+13)	1.59E+12 (2.56E+13)	-1.44E+11 (2.39E+12)	0.0141 (0.0140)	0.0205 (0.0167)	0.0231 (0.0186)	0.0247 (0.0194)	0.0281 (0.0225)	0.0386 (0.0463)	
	-0.25	-0.4780 (0.4192)	-2.45E+12 (1.74E+13)	-2.87E+12 (1.91E+13)	2.31E+12 (1.64E+13)	-2.09E+11 (1.73E+12)	0.0075 (0.0124)	0.0123 (0.0167)	0.0138 (0.0186)	0.0148 (0.0193)	0.0163 (0.0220)	0.0263 (0.0578)	
	0.00	-0.1594 (0.3896)	8.49E+01 (5.99E+02)	-1.88E+02 (9.78E+02)	6.93E+01 (3.47E+02)	-1.36E+02 (8.92E+02)	0.0203 (0.0146)	0.0288 (0.0141)	0.0317 (0.0153)	5.35E+11 (1.69E+13)	0.0366 (0.0178)	1.34E+13 (4.23E+14)	
	0.25	-0.0003 (0.2250)	0.0042 (0.4652)	-0.0036 (0.0241)	-0.0017 (0.2268)	-0.8362 (0.0528)	0.0233 (0.0130)	0.0349 (0.0042)	0.0387 (0.0029)	0.0400 (6.60E-10)	0.0453 (0.0038)	0.0529 (0.0133)	
	0.50	0.0032 (0.2317)	0.0119 (0.4597)	-0.0049 (0.0264)	-0.0062 (0.1966)	-0.7097 (0.0692)	0.0227 (0.0134)	0.0351 (0.0044)	0.0388 (0.0029)	0.0400 (3.11E-10)	0.0452 (0.0037)	0.0525 (0.0140)	
	0.75	0.0036 (0.2300)	0.0150 (0.4732)	-0.0033 (0.0235)	-0.0049 (0.1815)	-0.6222 (0.0921)	0.0230 (0.0136)	0.0350 (0.0048)	0.0389 (0.0028)	0.0400 (1.63E-10)	0.0453 (0.0039)	0.0523 (0.0143)	
	1.00	0.0018 (0.2423)	0.0153 (0.5024)	-0.0046 (0.0329)	-0.0034 (0.1842)	-0.5637 (0.1226)	0.0221 (0.0137)	0.0347 (0.0052)	0.0390 (0.0028)	0.0400 (6.90E-11)	0.0456 (0.0044)	0.0527 (0.0150)	

4. Simulation

Results ($n = 50$)

DGP	γ	$\hat{\theta}$	Lagrange multipliers			δ	Minimum	Q1	Median	π	Mean	Q3	Maximum
			λ_1	λ_2	λ_3								
t_5	-1.00	-0.0020 (0.1906)	0.0018 (0.2734)	-0.0009 (0.0112)	-1.50E-05 (0.0843)	-1.00E+04 (3.15E-06)	0.0101 (0.0057)	0.0180 (0.0016)	0.0195 (0.0009)	0.0200 (1.31E-11)	0.0219 (0.0015)	0.0219 (0.0086)	0.0306 (0.0298)
	-0.75	-0.0034 (0.1898)	-1.10E+05 (3.46E+06)	-9.25E+04 (2.92E+06)	5.64E+04 (1.78E+06)	-1.87E+03 (5.90E+04)	0.0096 (0.0059)	0.0182 (0.0017)	0.0196 (0.0012)	0.0200 (0.0006)	0.0218 (0.0017)	0.0218 (0.0130)	0.0298 (0.0269)
	-0.50	-0.0730 (0.3283)	-1.09E+12 (2.26E+13)	-1.09E+12 (2.49E+13)	4.96E+11 (1.05E+13)	-1.45E+11 (3.10E+12)	0.0079 (0.0062)	0.0169 (0.0051)	0.0182 (0.0053)	0.0185 (0.0053)	0.0202 (0.0060)	0.0202 (0.0102)	0.0269 (0.0102)
	-0.25	-0.3466 (0.5162)	-3.81E+12 (5.27E+13)	-2.51E+12 (3.10E+13)	1.38E+12 (1.57E+13)	-2.32E+11 (3.10E+12)	0.0054 (0.0064)	0.0122 (0.0089)	0.0130 (0.0095)	0.0131 (0.0095)	0.0141 (0.0103)	0.0141 (0.0214)	0.0195 (0.0214)
	0.00	-0.0412 (0.2638)	-0.2243 (16.4336)	-1.1852 (12.3150)	1.0296 (7.5537)	-1.2787 (10.2533)	0.0083 (0.0065)	0.0181 (0.0038)	0.0192 (0.0039)	0.0192 (0.0038)	0.0207 (0.0043)	0.0207 (0.0076)	0.0257 (0.0076)
	0.25	-0.0267 (0.2910)	0.0396 (0.2994)	-0.0063 (0.0276)	-0.0049 (0.0613)	-0.8422 (0.1181)	0.0082 (0.0067)	0.0183 (0.0025)	0.0196 (0.0017)	0.0200 (6.68E-11)	0.0216 (0.0015)	0.0216 (0.0228)	0.0304 (0.0228)
	0.50	-0.0380 (0.3263)	0.0395 (0.3666)	-0.0088 (0.0354)	-0.0029 (0.0601)	-0.7349 (0.2089)	0.0081 (0.0069)	0.0180 (0.0032)	0.0194 (0.0021)	0.0200 (3.04E-11)	0.0217 (0.0016)	0.0217 (0.0253)	0.0313 (0.0253)
	0.75	-0.0427 (0.3374)	0.0445 (0.4073)	-0.0101 (0.0435)	-0.0015 (0.0626)	-0.6584 (0.2283)	0.0176 (0.0070)	0.0194 (0.0037)	0.0200 (0.0020)	0.0220 (1.71E-11)	0.0307 (0.0021)	0.0307 (0.0197)	0.0070 (0.0197)
	1.00	-0.0565 (0.3611)	0.0677 (0.4643)	-0.0158 (0.0696)	-0.0019 (0.0629)	-0.6172 (0.2600)	0.0172 (0.0071)	0.0194 (0.0042)	0.0200 (0.0019)	0.0225 (8.12E-11)	0.0310 (0.0028)	0.0310 (0.0174)	0.0071 (0.0174)
Normal	-1.00	-0.0060 (0.1478)	-0.0155 (0.3186)	4.60E-05 (0.0038)	0.0059 (0.1322)	-1.00E+04 (1.95E-06)	0.0134 (0.0046)	0.0179 (0.0015)	0.0194 (0.0009)	0.0200 (9.24E-13)	0.0220 (0.0013)	0.0220 (0.0061)	0.0272 (0.0061)
	-0.75	-0.0071 (0.1602)	-3.23E+06 (5.49E+07)	-2.92E+06 (5.06E+07)	2.01E+06 (3.50E+07)	-8.57E+04 (1.48E+06)	0.0130 (0.0050)	0.0178 (0.0021)	0.0193 (0.0018)	0.0199 (0.0017)	0.0219 (0.0023)	0.0219 (0.0062)	0.0268 (0.0062)
	-0.50	-0.1240 (0.3453)	-1.57E+12 (4.53E+13)	-1.34E+12 (3.84E+13)	1.02E+12 (2.96E+13)	-7.21E+10 (1.97E+12)	0.0105 (0.0066)	0.0151 (0.0067)	0.0164 (0.0072)	0.0168 (0.0073)	0.0186 (0.0082)	0.0186 (0.0175)	0.0231 (0.0175)
	-0.25	-0.4980 (0.4615)	-2.33E+12 (1.85E+13)	-2.54E+12 (2.07E+13)	1.93E+12 (1.57E+13)	-1.89E+11 (2.01E+12)	0.0047 (0.0066)	0.0076 (0.0090)	0.0083 (0.0097)	0.0084 (0.0099)	0.0093 (0.0109)	0.0093 (0.0135)	0.0117 (0.0135)
	0.00	-0.0493 (0.2509)	0.5854 (110.3248)	-12.4907 (182.4158)	7.1143 (57.5511)	-4.4559 (150.0305)	0.0117 (0.0062)	0.0172 (0.0045)	0.0185 (0.0047)	0.0188 (0.0047)	0.0206 (0.0053)	0.0206 (0.0074)	0.0241 (0.0074)
	0.25	-0.0017 (0.1491)	-0.0130 (0.2887)	-0.0007 (0.0077)	0.0062 (0.1187)	-0.8165 (0.0243)	0.0120 (0.0062)	0.0183 (0.0014)	0.0197 (0.0007)	0.0200 (9.23E-11)	0.0219 (0.0014)	0.0219 (0.0044)	0.0253 (0.0044)
	0.50	-0.0021 (0.1480)	-0.0094 (0.2914)	-0.0028 (0.0162)	0.0050 (0.1112)	-0.6869 (0.0342)	0.0117 (0.0065)	0.0183 (0.0016)	0.0198 (0.0008)	0.0200 (4.47E-11)	0.0219 (0.0015)	0.0219 (0.0045)	0.0251 (0.0045)
	0.75	0.0013 (0.1563)	-0.0018 (0.3178)	-0.0042 (0.0249)	0.0019 (0.1080)	-0.5990 (0.0591)	0.0114 (0.0067)	0.0182 (0.0019)	0.0197 (0.0009)	0.0200 (4.64E-11)	0.0220 (0.0017)	0.0220 (0.0057)	0.0252 (0.0057)
	1.00	-0.0089 (0.1661)	0.0166 (0.3275)	-0.0063 (0.0460)	-0.0025 (0.1001)	-0.5346 (0.0924)	0.0111 (0.0068)	0.0181 (0.0021)	0.0198 (0.0008)	0.0200 (3.36E-11)	0.0220 (0.0018)	0.0220 (0.0070)	0.0252 (0.0070)

Remarks

- γ should be viewed as a learning hyperparameter.
 - ▶ It determines the curvature of the CRPD loss.
 - ▶ Role in asymptotics
 - ★ *First order*: the asymptotic distribution of $\hat{\theta}$ does not depend on γ .
 - ★ *Second order*: γ enters the higher-order terms of both the structural estimator and the Lagrange multipliers.
 - ★ Hence differences across γ appear through bias, coverage distortion, and other finite-sample effects, not through first-order efficiency.
- Implication
 - ▶ Choosing γ means choosing the loss curvature that delivers the best finite-sample performance.
 - ▶ Given this interpretation, a data-driven way such as cross-validation is a natural way to select γ , since it targets finite-sample risk.
- Caveat
 - ▶ γ should be selected using a common out-of-sample criterion.
 - ▶ Thus, minimized CRPD values are not directly comparable across γ , because each γ defines a different loss.

Algorithm 1 CRPD Estimation with Cross-Validated Hyperparameter γ

Require: Data $\{Z_i\}_{i=1}^n$, moment function $m(Z, \theta)$, candidate set Γ , folds K

Ensure: Selected $\hat{\gamma}$ and estimates $(\hat{\theta}, \hat{\lambda}, \hat{\delta})$

- 1: Randomly partition $\{1, \dots, n\}$ into K folds $\{\mathcal{F}_k\}_{k=1}^K$.
- 2: **for** $\gamma \in \Gamma$ **do**
- 3: Set $\text{CV}(\gamma) \leftarrow 0$.
- 4: **for** $k = 1, \dots, K$ **do**
- 5: Training set $\mathcal{T}_k = \{1, \dots, n\} \setminus \mathcal{F}_k$; validation set $\mathcal{V}_k = \mathcal{F}_k$.
- 6: Fit CRPD on $\mathcal{T}_k$ at fixed γ :

$$(\hat{\theta}_k(\gamma), \hat{\lambda}_k(\gamma), \hat{\delta}_k(\gamma)) \in \arg \min_{\theta, \lambda, \delta} \mathcal{J}_\gamma(\pi(\lambda, \delta, \theta), \mathbf{i}/|\mathcal{T}_k|)$$

subject to

$$\sum_{i \in \mathcal{T}_k} \pi_i = 1, \quad \sum_{i \in \mathcal{T}_k} \pi_i m(Z_i, \theta) = 0.$$

- 7: Compute out-of-sample moment instability on $\mathcal{V}_k$:

$$\mathcal{V}_k(\gamma) = \left\| \frac{1}{|\mathcal{V}_k|} \sum_{i \in \mathcal{V}_k} m(Z_i, \hat{\theta}_k(\gamma)) \right\|^2.$$

- 8: Update $\text{CV}(\gamma) \leftarrow \text{CV}(\gamma) + \mathcal{V}_k(\gamma)/K$.

- 9: **end for**

- 10: **end for**

- 11: Select $\hat{\gamma} \in \arg \min_{\gamma \in \Gamma} \text{CV}(\gamma)$.

- 12: Refit on the full sample at $\hat{\gamma}$:

$$(\hat{\theta}, \hat{\lambda}, \hat{\delta}) \in \arg \min_{\theta, \lambda, \delta} \mathcal{J}_{\hat{\gamma}}(\pi(\lambda, \delta, \theta), \mathbf{i}/n)$$

subject to

$$\sum_{i=1}^n \pi_i = 1, \quad \sum_{i=1}^n \pi_i m(Z_i, \theta) = 0.$$

Empirical Application

5. Empirical Application

Data

- We revisit Owen (2001, Table 3.2, p.44), which reports the pounds of milk produced and the number of days milked in 1936 for $n = 22$ dairy cows.

The Pounds of Milk Produced and the Number of Days Milked (Owen, 2001)

Cow	Milk	Days	Milk per day
Belle	4,628	248	18.6613
Bessie	3,914	312	12.5449
Beauty	2,851	278	10.2554
Bertha	2,765	267	10.3558
Blossom	2,067	235	8.7957
Blackie	1,817	211	8.6114
Buttercup	4,248	327	12.9908
Dolly	3,037	262	11.5916
Dandy	2,627	235	11.1787
Irene	3,527	255	13.8314
Iona	3,257	260	12.5269
Jessie	2,793	283	9.8693
Ladybird	1,932	256	7.5469
Marigold	2,992	254	11.7795
Peggy	4,444	268	16.5821
Pansy	3,944	333	11.8438
Queen Ann	5,448	299	18.2207
Queeny	4,512	298	15.1409
Rosemary	5,435	317	17.1451
Rosy	4,792	415	11.5470
Sally	2,635	264	9.9811
Wildy	3,346	271	12.3469

Note: Owen (2001) states: "Shown are the pounds of milk produced and the number of days milked, in 1936, for 22 dairy cows. It is not known whether Rosy's number of days milked is in error, or if it includes some days from 1935."

5. Empirical Application

Motivation

- Suppose we want to estimate the mean daily milk yield θ .

5. Empirical Application

Motivation

- Suppose we want to estimate the mean daily milk yield θ .
- Overall, the distribution of milk per day (mpd) reflects heterogeneity in production levels across cows.
 - ▶ The average milk production per day is 12.43 units, with a standard deviation of 3.08, indicating moderate dispersion across cows.
 - ▶ The minimum and maximum values are 7.55 and 18.66, respectively, suggesting substantial variation in daily milk productivity within the sample.

Statistic	n	Mean	St. Dev.	Minimum	Maximum
mpd	22	12.4250	3.0750	7.5470	18.6610

Note: mpd denotes milk per day (lb/day). n is the number of observations. St. Dev. denotes the standard deviation.

5. Empirical Application

Motivation

- The plug-in estimator

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

assigns identical weight to every observation.

- ▶ In small samples (e.g., $n = 22$), a single atypical record can distort the sample moment and heavily influence the estimate.

5. Empirical Application

Motivation

- The plug-in estimator

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

assigns identical weight to every observation.

- ▶ In small samples (e.g., $n = 22$), a single atypical record can distort the sample moment and heavily influence the estimate.
- The CRPD estimator is data-driven:
 - ▶ selects the loss curvature through γ ,
 - ▶ enforces the moment condition through endogenous reweighting of the data,
 - ★ observations generating large moment violations receive smaller weights,
 - ★ observations consistent with the moment condition receive relatively larger weights,
 - ▶ thereby satisfies the moment condition while limiting the influence of atypical observations.

5. Empirical Application

Empirical specification

Let $X_i = mpd_i$ denote milk per day (lb/day) for cow i .

- Parameter of interest

- ▶ Population mean: $\mu = E[X_i]$.
- ▶ Identifying moment: $E[X_i - \mu] = 0$.
- ▶ Issue: with a single moment, CRPD collapses to the trivial uniform-weight solution ($\hat{\mu} = \bar{X}$).

- Instrument moment for overidentification

- ▶ Use observable variable $Days_i$:

$$E[(X_i - \mu) Days_i] = 0.$$

- ▶ Pearson correlation test: $\rho = 0.32$ (no significant correlation).

- Thus, we have the following moment system:

$$g(X_i, \mu) = \begin{pmatrix} X_i - \mu \\ (X_i - \mu) Days_i \end{pmatrix}.$$

5. Empirical Application

Results

We implement 5-fold cross-validation over $\gamma \in [-10, 10]$ (step 0.001).

Mode	γ	$\hat{\theta}$
Pre-specified	-2	11.5790
Pre-specified	-1.5	11.0546
Pre-specified	-1	12.8245
Pre-specified	-0.5	12.8245
Pre-specified	0	12.8245
Pre-specified	0.5	12.8245
Pre-specified	1	12.8245
Cross-validated	-0.654	12.8245

- Cross-validation selects $\hat{\gamma} = -0.654$, yielding $\hat{\theta} = 12.8245$ lb/day.
- Choosing γ a priori (e.g., $\gamma = -2, -1.5$) can lead to materially different estimates,
 - ▶ in some cases performing worse than the simple mean $\bar{X} = 12.4250$.
- For moderate values of γ , the estimates remain stable (up to the fourth decimals),
 - ▶ the data are broadly consistent with the moment restrictions without requiring excessive reweighting.

5. Empirical Application

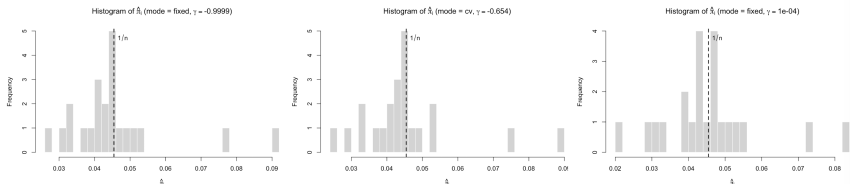
Variation in estimated weights

- The weights $\hat{\pi}(\gamma) = \{\hat{\pi}_i(\gamma)\}_{i=1}^n$ provide diagnostics: $\hat{\theta}(\gamma) = \sum_{i=1}^n \hat{\pi}_i(\gamma) X_i$.
 - ▶ More dispersed weights indicate reliance on a subset of observations.
 - ▶ Large weights highlight observations that strongly influence the moments.
 - ▶ Clusters of similar weights may suggest heterogeneity or latent subgroups.

5. Empirical Application

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- $\hat{\theta}$ is nearly invariant across moderate γ (up to the fourth decimal), but $\hat{\pi}(\gamma)$ differ.
 - ▶ $\gamma \approx 0$ (ET): smooth weights close to the uniform benchmark $1/n$.
 - ▶ More negative γ (EL-like): allows stronger reweighting, producing more dispersed weights and greater influence of a subset of observations.
 - ▶ $\hat{\gamma} = -0.654$ lies between these regimes, yielding an intermediate pattern.

Conclusion

Conclusion

- **Moment-based estimation in the era of big data**
 - ▶ High dimensionality (p large relative to n) makes finite-sample properties central.
 - ▶ Conventional GMM can become unstable or perform poorly.
 - ▶ CRPD — the dual formulation of GEL — provides a useful alternative.
- **CRPD from a machine learning perspective**
 - ▶ The power parameter γ indexes a family of estimators.
 - ▶ Popular benchmark choices: $\gamma = 0$ (ET) and $\gamma = -1$ (EL).
- **The power parameter γ as a learning hyperparameter**
 - ▶ γ determines the curvature of the loss function.
 - ▶ Finite-sample performance of both structural parameters and Lagrange multipliers depends on γ .
 - ▶ The optimal γ can be selected data-adaptively (e.g., via cross-validation).
- **Bridging econometrics and machine learning**
 - ▶ Interpret classical statistical objects through the lens of modern learning theory.

Future research

- γ may effectively behave like a structural parameter under moment misspecification.
 - ▶ ET and EL are no longer first-order equivalent (Schennach, 2007).
 - ▶ Different values of γ lead to different pseudo-true parameters.
 - ▶ Thus, the choice of γ affects the population estimand.

Appendix

γ value	Core loss term	Geometry	Interpretation
$\gamma = -1$ (EL)	$-\sum_i \log(n\pi_i)$	Log barrier	Empirical likelihood. The objective diverges as $\pi_i \downarrow 0$, strongly discouraging vanishing weights and enforcing interior solutions.
$\gamma = 0$ (ET)	$\sum_i \pi_i \log(n\pi_i)$	Logarithmic	Exponential tilting. Corresponds to the Kullback–Leibler divergence from the uniform distribution; weights are smoothly reweighted and remain close to $1/n$ when the moment conditions are nearly satisfied.
$\gamma = 1$ (Pearson χ^2)	$\sum_i \pi_i^2$	Quadratic	Pearson χ^2 divergence. Equivalent to the squared Euclidean distance from the uniform distribution on the probability simplex; in moment space it corresponds to a quadratic (Hotelling-type) distance.

Theorem 2 (Glivenko–Cantelli)

Let $\{X_i\}_{i=1}^{\infty}$ be an i.i.d. sequence of random variables with distribution function F_0 . Define the empirical distribution function

$$\hat{F}_n(x) \equiv \frac{1}{n} \sum_{i=1}^n 1\{X_i \leq x\}.$$

Then,

$$\sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F_0(x)| \xrightarrow{a.s.} 0 \quad \text{as } n \rightarrow \infty.$$

- By the Glivenko–Cantelli theorem, the empirical distribution—which assigns uniform mass to each realized sample point—converges uniformly to the true distribution.
- Hence, the uniform distribution over the sample points (not over the population support) is a consistent estimator of the true distribution.

What does γ do? [▶ Back](#)

$$\text{CRPD criterion: } l_\gamma(\mathbf{p}, \mathbf{q}) = \frac{1}{\gamma(\gamma+1)} \sum_{i=1}^n p_i \left[\left(\frac{p_i}{q_i} \right)^\gamma - 1 \right].$$

- The key transformation is:

$$r_j^\gamma = \exp(\gamma \log r_j), \quad r_j \equiv \frac{p_j}{q_j}.$$

- ▶ If $r_j > 1$ (large positive deviation), r_j^γ increases in γ .
- ▶ If $r_j < 1$ (large negative deviation), r_j^γ decreases in γ .
- ▶ Thus, γ governs how strongly probability ratios are magnified or compressed, i.e., how sensitively extreme discrepancies affect the divergence.
- ET ($\gamma = 0$) vs. EL ($\gamma = -1$)
 - ▶ Positive γ : power amplification of large ratios.
 - ▶ Negative γ : reciprocal-type transformation that shrinks extreme deviations.
 - ▶ ET: deviations enter logarithmically (moderate, smooth amplification).
 - ▶ EL: deviations enter reciprocally (r^{-1}), inducing stronger local down-weighting of extremes.
 - ▶ Remark. ET and EL differ in their weighting mechanisms. ET tilts weights smoothly via an exponential form, while EL employs a reciprocal structure that reacts sharply to residuals near the feasibility boundary.

Why exact identification leads to uniform weights

[▶ Back](#)

- Suppose the model is **exactly identified**

$$q = p$$

- Then we can find $\hat{\theta}$ such that the sample moment condition holds:

$$\bar{g}(\hat{\theta}) = \frac{1}{n} \sum_{i=1}^n g_i(\hat{\theta}) = 0$$

- The moment constraint in CRPD estimation is

$$\sum_{i=1}^n \pi_i g_i(\theta) = 0.$$

- If $\theta = \hat{\theta}$ satisfies $\bar{g}(\hat{\theta}) = 0$, then this constraint is satisfied with

$$\pi_i = \frac{1}{n}, \quad i = 1, \dots, n.$$

- Therefore

- ▶ the Lagrange multiplier $\lambda = 0$
- ▶ the CRPD objective is minimized at the empirical distribution

- **Conclusion**

- ▶ exact identification $\Rightarrow$ no reweighting needed